HeatWatch: Predicting Delays in NYC Heating and Hot Water Complaints

AI for the SDGs • Assignment 1 Concept Note
Group 11: Mert Sürmeli, Mehmet Batu Solar, Abdurrahman Dıraz | Yıldız Technical University

Overview & SDG connection

Tenants in New York City can report a loss of heat or hot water to 311. Our project investigates whether information available at the time of the complaint can predict if the service-request record will take more than 72 hours to close. We link the question to UN SDG 11, Target 11.1 (adequate housing and basic services). Our tool is an academic early-warning prototype, not a real service-prioritisation system. The 72-hour mark is our study threshold, not a government deadline.

Research questions

1. How do complaint counts and closure times change across months? 2. What share of closed requests exceeds 72 hours? 3. Does this vary across boroughs? 4. Can the first models rank later-month delays above a simple baseline?

Public dataset & preparation

We queried the official NYC 311 Service Requests from 2020 to Present dataset (ID: erm2-nwe9) for HEAT/HOT WATER requests opened from 1 October 2024 to 31 May 2025. Fields used: request ID, created/closed timestamps, complaint detail and borough; we also derived month, weekday and hour. No resident names, addresses or coordinates enter the model. We deduplicated request IDs, parsed dates and calculated closed date − created date in hours. Missing closure dates were excluded from supervised training and reported separately.

Requests retrieved	Missing close date	Closed records	Delayed >72h
280,427	4,241 (1.51%)	276,186	48,695 (17.63%)

Median administrative closure time: 42 hours. In this report, “closed” is an administrative status and is not proof that the heating problem was fixed.

First model: input → algorithm → output

Input at opening	Method	Model output
Borough, complaint detail, month, weekday, hour	Logistic regression, random forest, simple baseline	Estimated risk of record closure taking >72 h

The model sees only information available when the complaint is made. It does not receive the closure date, closure time, final status or resolution text as input. The program is implemented in Python with pandas and scikit-learn.

Evaluation design

Training period: October 2024–February 2025 (226,910 valid closed records). Held-out test: March–May 2025 (49,276). We compare a prior-only baseline with two learning methods using ROC-AUC, PR-AUC and Brier score. This is a chronological, not random, split.

Observed EDA findings

The number of requests was greatest in January 2025. Across records with closure dates, Manhattan had a 27.1% over-72-hour rate versus 9.8% in the Bronx. These are descriptive differences; the dataset does not show that borough itself causes administrative delays.

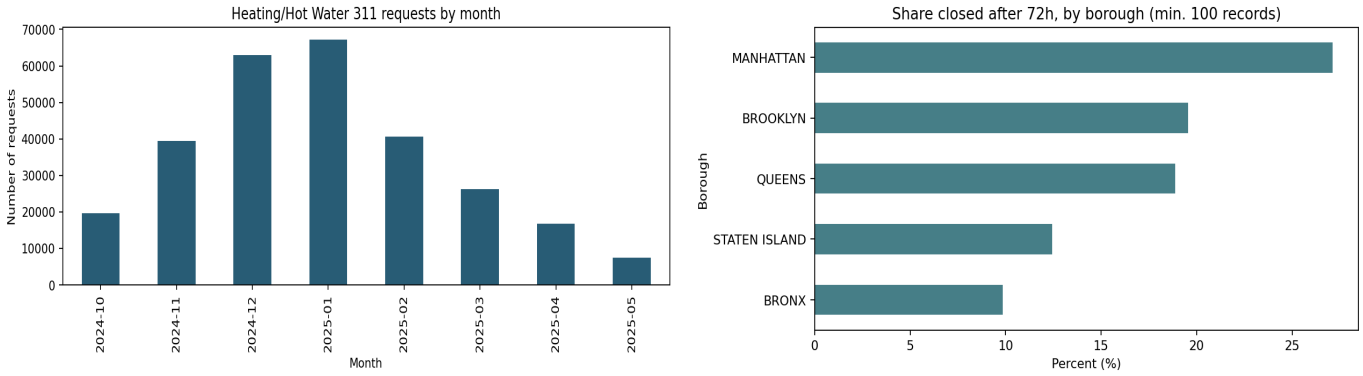


Figure 1. Requests by month (left, including missing closure dates); delay share by borough (right, closed records only).

Initial model results (held-out March–May period)

Method	ROC-AUC ↑	PR-AUC ↑	Brier ↓
Prior-only baseline	0.500	0.052	0.0722
Logistic regression	0.687	0.128	0.0736
Random forest	0.697	0.124	0.0686

Both machine-learning methods rank delayed cases above the constant baseline. Logistic regression has the highest PR-AUC (0.128); random forest has slightly higher ROC-AUC (0.697) and lower Brier score (0.0686). No single model leads on all metrics. However, at a 0.50 probability threshold random forest identifies none of the delayed test cases, so this is not yet a usable alert system.

A crucial limitation

The delayed-case rate falls from 20.33% in the training period to 5.21% in the test period. The models have difficulty with this seasonal change. Test months were also unseen categories for the month feature. In the next iteration we should revise the time features, tune and validate thresholds, and test across multiple heating seasons.

Ethics, deliverables & next phase

We excluded 4,241 requests without closure dates, which may bias the measured rates. Complaint records cover reported problems, not every affected household. We will not use predictions to reject or deprioritise services, infer repair success, or generalise NYC results to Türkiye. Next: implement a small Streamlit demonstration that displays a probability with this caveat, publish reproducible code, and conduct stronger testing.

Sources

NYC Open Data, 311 Service Requests (erm2-nwe9): https://data.cityofnewyork.us/d/erm2-nwe9
NYC HPD, Heat and Hot Water Information: https://www.nyc.gov/site/hpd/services-and-information/heat-and-hot-water-information.page
United Nations, SDG 11.1: https://sdgs.un.org/goals/goal11